Chapter 9

Biomolecules

There is a wide diversity in living organisms in our biosphere. Are all living organisms made of the same chemicals, that is the same elements and compounds?

Living against non-living

- elemental analysis of a plant tissue, animal tissue or microbial paste gives a list of elements such as carbon, hydrogen and oxygen, with their content per unit mass
- the same analysis on a piece of earth's crust gives a similar list, and in absolute terms no differences can be made out, since every element in the crust is also present in living tissue
- a closer examination reveals that the relative abundance of carbon and hydrogen against other elements is higher in any living organism than in earth's crust

Two fractions from a tissue

- take any living tissue, a vegetable or a piece of liver, and grind it in trichloroacetic acid using a mortar and pestle to obtain a thick slurry
- strained through cheesecloth or cotton, it gives two fractions
- one is the filtrate, technically the acid-soluble pool, and the second the retentate or acid-insoluble fraction
- scientists have found thousands of organic compounds in the acid-soluble pool

In this chapter

9.1 Analysing chemical composition

9.3 Amino acids, lipids and bases

9.7 Biomacromolecules

9.11 Structure of proteins

9.12 Enzymes

9.17 Classification of enzymes

9.1 Isolating and identifying a compound

How a compound is separated

- one extracts the compounds, then subjects the extract to various separation techniques until a compound has been separated from all others, that is isolated and purified
- analytical techniques applied to the compound give an idea of its molecular formula and probable structure
- all the carbon compounds obtained from living tissues can be called biomolecules

Finding the inorganic part

- living organisms also have inorganic elements and compounds, shown by a destructive experiment
- a small amount of living tissue is weighed, this being the wet weight, and dried until all the water evaporates, the remaining material giving the dry weight
- if the tissue is then fully burnt, all carbon compounds are oxidised to gaseous form and removed, and what remains is called ash, containing inorganic elements such as calcium and magnesium
- inorganic compounds such as sulphate and phosphate are also seen in the acid-soluble fraction

What the two analyses give

- elemental analysis gives the elemental composition of living tissues, as hydrogen, oxygen, chlorine and carbon
- analysis for compounds gives an idea of the organic and inorganic constituents present
- from a chemistry point of view one can identify functional groups such as aldehydes, ketones and aromatic compounds, but from a biological point of view they are classified into amino acids, nucleotide bases and fatty acids

Table 9.A A Comparison of Elements Present in Non-living and Living Matter*

Element	% Weight of Earth's crust	% Weight of Human body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

* Adapted from CNR Rao, Understanding Chemistry, Universities Press, Hyderabad.

Table 9.B A List of Representative Inorganic Constituents of Living Tissues

Component	Formula
Sodium	Na+
Potassium	K+
Calcium	Ca++
Magnesium	Mg++
Water	H2O
Compounds	NaCl, CaCO3, PO4 3–, SO4 2–

9.2 Amino acids

Substituted methanes

- amino acids are organic compounds containing an amino group and an acidic group as substituents on the same carbon, the alpha carbon, hence they are called alpha amino acids
- they are substituted methanes, with four substituent groups occupying the four valency positions, namely hydrogen, carboxyl group, amino group and a variable R group
- based on the nature of the R group there are many amino acids, but those occurring in proteins are only of twenty types
- the R group could be a hydrogen, giving glycine, a methyl group giving alanine, or hydroxy methyl giving serine

Their properties

- the chemical and physical properties are essentially those of the amino, carboxyl and R functional groups
- by the number of amino and carboxyl groups there are acidic amino acids such as glutamic acid, basic such as lysine, and neutral such as valine
- there are also aromatic amino acids, tyrosine, phenylalanine and tryptophan
- a particular property is the ionizable nature of the amino and carboxyl groups, so that in solutions of different pH the structure of amino acids changes, one form being called the zwitterionic form

9.3 Lipids

Fatty acids and glycerides

- lipids are generally water insoluble, and could be simple fatty acids, a fatty acid having a carboxyl group attached to an R group
- the R group could be a methyl, or ethyl, or a higher number of methylene groups, from 1 carbon to 19 carbons, palmitic acid has 16 carbons including the carboxyl carbon and arachidonic acid 20
- fatty acids may be saturated, without a double bond, or unsaturated, with one or more carbon to carbon double bonds
- another simple lipid is glycerol, which is trihydroxy propane, and many lipids have both glycerol and fatty acids, the acids being esterified with glycerol to give monoglycerides, diglycerides and triglycerides

Fats, oils and phospholipids

- these are also called fats and oils based on melting point, oils having the lower melting point, as in gingelly oil, and so remaining oil in winters
- some lipids have phosphorous and a phosphorylated organic compound in them, and these are phospholipids, found in cell membrane, lecithin being one example
- some tissues, especially the neural tissues, have lipids with more complex structures

9.4 Nitrogen bases and nucleotides

From base to nucleotide

- living organisms have a number of carbon compounds in which heterocyclic rings are found, some of these being the nitrogen bases adenine, guanine, cytosine, uracil and thymine
- when found attached to a sugar they are called nucleosides, and if a phosphate group is also esterified to the sugar they are nucleotides
- adenosine, guanosine, thymidine, uridine and cytidine are nucleosides, while adenylic acid, thymidylic acid, guanylic acid, uridylic acid and cytidylic acid are nucleotides
- nucleic acids such as DNA and RNA consist of nucleotides only, and function as genetic material

Sugars and amino acids

- Glucose, C6H12O6, a six carbon sugar ring
- Ribose, C5H10O5, a five carbon sugar ring
- Glycine, alanine and serine, three of the twenty amino acids

Fats and oils

- Palmitic acid, a fatty acid
- Glycerol, and a triglyceride whose three R groups are fatty acids
- Phospholipid, shown as lecithin
- Cholesterol

Bases and nucleotides

- Adenine, a purine, uracil, a pyrimidine
- Adenosine and uridine, the corresponding nucleosides
- Adenylic acid, a nucleotide, carrying a phosphate on the sugar

Figure 9.A Diagrammatic representation of small molecular weight organic compounds in living tissues

9.5 Primary and secondary metabolites

Two kinds of metabolite

- the most exciting aspect of chemistry deals with isolating thousands of compounds, small and big, from living organisms, determining their structure and if possible synthesising them
- a list of biomolecules would have thousands of organic compounds including amino acids and sugars, and these are called metabolites
- in animal tissues one notices all such categories of compounds, and these are primary metabolites
- in plant, fungal and microbial cells one sees thousands of compounds other than these, such as alkaloids, flavonoids, rubber, essential oils, antibiotics, coloured pigments, scents, gums and spices, called secondary metabolites

What they are for

- primary metabolites have identifiable functions and play known roles in normal physiological processes
- we do not at the moment understand the role or functions of all the secondary metabolites in host organisms
- many of them are nevertheless useful to human welfare, such as rubber, drugs, spices, scents and pigments, and some have ecological importance

9.6 Biomacromolecules

Two size classes

- one feature is common to all compounds found in the acid soluble pool, that their molecular weights range from 18 to around 800 daltons
- the acid insoluble fraction has only four types of organic compounds, proteins, nucleic acids, polysaccharides and lipids, and with the exception of lipids these have molecular weights in the range of ten thousand daltons and above
- so biomolecules are of two types: those under one thousand dalton, usually referred to as micromolecules or simply biomolecules, and those in the acid insoluble fraction, called macromolecules or biomacromolecules
- the molecules in the insoluble fraction, lipids excepted, are polymeric substances

Table 9.C Some Secondary Metabolites

Class	Examples
Pigments	Carotenoids, Anthocyanins, etc.
Alkaloids	Morphine, Codeine, etc.
Terpenoides	Monoterpenes, Diterpenes etc.
Essential oils	Lemon grass oil, etc.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, etc.
Polymeric substances	Rubber, gums, cellulose

Why lipids sit in the macromolecular fraction

- lipids are indeed small molecular weight compounds, present not only as such but also arranged into structures such as cell membrane and other membranes
- grinding a tissue disrupts the cell structure, and the membranes break into pieces forming vesicles which are not water soluble
- these membrane fragments as vesicles separate along with the acid insoluble pool and hence fall in the macromolecular fraction, so lipids are not strictly macromolecules

What the two pools represent

- the acid soluble pool represents roughly the cytoplasmic composition
- the macromolecules from cytoplasm and organelles become the acid insoluble fraction
- together they represent the entire chemical composition of living tissues or organisms
- arranged class-wise by abundance, water is the most abundant chemical in living organisms

9.7 Proteins

Polypeptides

- proteins are polypeptides, linear chains of amino acids linked by peptide bonds
- each protein is a polymer of amino acids, and as there are 20 types of amino acids a protein is a heteropolymer and not a homopolymer, a homopolymer having only one type of monomer repeating n number of times
- certain amino acids are essential for our health and have to be supplied through diet, so dietary proteins are the source of essential amino acids
- amino acids can therefore be essential or non-essential, the latter being those our body can make while essential ones come through our food

Table 9.D Average Composition of Cells

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

Table 9.E Some Proteins and their Functions

Protein	Functions
Collagen	Intercellular ground substance
Trypsin	Enzyme
Insulin	Hormone
Antibody	Fights infectious agents
Receptor	Sensory reception (smell, taste, hormone, etc.)
GLUT-4	Enables glucose transport into cells

A branched chain of glucose units

The right end is the reducing end

The left end is the non-reducing end

Branches arise along the chain

Figure 9.B Diagrammatic representation of a portion of glycogen

What proteins do

- proteins carry out many functions in living organisms: some transport nutrients across cell membrane, some fight infectious organisms, some are hormones and some are enzymes
- collagen is the most abundant protein in the animal world
- Ribulose bisphosphate Carboxylase-Oxygenase, RuBisCO, is the most abundant protein in the whole of the biosphere

9.8 Polysaccharides

Long chains of sugars

- the acid insoluble pellet also has polysaccharides as another class of macromolecules, which are threads containing different monosaccharides as building blocks
- cellulose is a polymeric polysaccharide consisting of only one type of monosaccharide, glucose, and is therefore a homopolymer
- starch is a variant of this, present as a store house of energy in plant tissues, while animals have another variant called glycogen, and inulin is a polymer of fructose
- in a polysaccharide chain the right end is called the reducing end and the left end the non-reducing end, and it has branches

Helices and iodine

- starch forms helical secondary structures, and can hold iodine molecules in the helical portion
- the starch and iodine complex is blue in colour
- cellulose does not contain complex helices and hence cannot hold iodine

More complex polysaccharides

- plant cell walls are made of cellulose, and paper made from plant pulp and cotton fibre is cellulosic
- more complex polysaccharides have as building blocks amino-sugars and chemically modified sugars, such as glucosamine and N-acetyl galactosamine
- exoskeletons of arthropods have a complex polysaccharide called chitin, and these complex polysaccharides are mostly homopolymers

9.9 Nucleic acids

Polynucleotides

- the other type of macromolecule in the acid insoluble fraction is the nucleic acid, and together with polysaccharides and polypeptides these comprise the true macromolecular fraction of any living tissue
- the building block is a nucleotide, which has three chemically distinct components, a heterocyclic compound, a monosaccharide, and a phosphoric acid or phosphate
- the heterocyclic compounds are the nitrogenous bases adenine, guanine, uracil, cytosine and thymine, adenine and guanine are substituted purines while the rest are substituted pyrimidines, the skeletal heterocyclic rings being called purine and pyrimidine
- the sugar is either ribose, a monosaccharide pentose, or deoxyribose, a nucleic acid containing deoxyribose is deoxyribonucleic acid and one containing ribose is ribonucleic acid

9.10 Structure of proteins

What structure means

- proteins are heteropolymers containing strings of amino acids, and structure means different things in different contexts
- in inorganic chemistry structure invariably refers to the molecular formulae, organic chemists write a two dimensional view, and physicists conjure up three dimensional views
- biologists describe the protein structure at four levels

Primary structure

- the sequence of amino acids, that is the positional information in a protein, which is the first amino acid, which the second, and so on, is called the primary structure
- a protein is imagined as a line, the left end represented by the first amino acid and the right end by the last

Primary, the polypeptide as a sequence

Secondary, the alpha helix and the beta pleated sheet

Tertiary, the chain folded upon itself, with hydrogen and disulphide bonds

Quaternary, several folded subunits arranged together

Figure 9.C Various levels of Protein Structure

The two terminals

- the first amino acid is also called the N-terminal amino acid
- the last amino acid is called the C-terminal amino acid

Secondary and tertiary

- a protein thread does not exist throughout as an extended rigid rod, the thread is folded in the form of a helix, similar to a revolving staircase
- only some portions of the thread are arranged as a helix, and in proteins only right handed helices are observed, other regions being folded into other forms, which is the secondary structure
- the long protein chain is also folded upon itself like a hollow woolen ball, giving the tertiary structure, which gives a three dimensional view
- tertiary structure is absolutely necessary for the many biological activities of proteins

Quaternary

- some proteins are an assembly of more than one polypeptide or subunit
- the manner in which these individual folded subunits are arranged with respect to each other, for instance as a linear string of spheres or spheres arranged one upon another as a cube or plate, is the architecture of a protein, called the quaternary structure
- adult human haemoglobin consists of 4 subunits, two identical to each other, so two subunits of alpha type and two of beta type together constitute human haemoglobin

9.11 Enzymes

Almost all enzymes are proteins

- there are some nucleic acids that behave like enzymes, and these are called ribozymes
- an enzyme, like any protein, has a primary structure, that is the amino acid sequence, and also the secondary and tertiary structure

The active site

- when the chain criss-crosses upon itself, many crevices or pockets are made, and one such pocket is the active site, into which the substrate fits
- thus enzymes, through their active site, catalyse reactions at a high rate
- enzyme catalysts differ from inorganic catalysts in many ways, but one major difference is that inorganic catalysts work efficiently at high temperatures and pressures while enzymes get damaged above about 40 degrees C
- enzymes isolated from organisms living under extremely high temperatures, as in hot vents and sulphur springs, are stable and retain catalytic power up to 80 to 90 degrees C, so thermal stability is an important quality of enzymes from thermophilic organisms

9.12 Chemical reactions

Physical and chemical change

- a physical change refers to a change in shape without breaking of bonds, and a change in state of matter, as when ice melts into water or water becomes vapour, is also physical
- when bonds are broken and new bonds formed during transformation, that is a chemical reaction
- an inorganic example is barium hydroxide with sulphuric acid giving barium sulphate and water, and hydrolysis of starch into glucose is an organic chemical reaction

Rate

- the rate of a physical or chemical process refers to the amount of product formed per unit time, and can be called velocity if the direction is specified
- rates are influenced by temperature among other factors, a general rule of thumb being that rate doubles or decreases by half for every 10 degrees C change in either direction
- catalysed reactions proceed at rates vastly higher than uncatalysed ones, as when carbon dioxide and water form carbonic acid under carbonic anhydrase

The power of enzymes

- without any enzyme the carbonic acid reaction is very slow, about 200 molecules being formed in an hour
- using carbonic anhydrase, present within the cytoplasm, the reaction speeds dramatically, with about 600,000 molecules formed every second
- the enzyme has accelerated the reaction rate by about 10 million times

Metabolic pathways

- there are thousands of types of enzymes, each catalysing a unique chemical or metabolic reaction
- a multistep chemical reaction, when each of the steps is catalysed by the same enzyme complex or by different enzymes, is called a metabolic pathway
- glucose becoming pyruvic acid is such a pathway, running through ten different enzyme catalysed metabolic reactions
- the same pathway with one or two additional reactions gives a variety of end products: lactic acid in skeletal muscle under anaerobic conditions, pyruvic acid under normal aerobic conditions, and ethanol in yeast during fermentation

9.13 How enzymes bring about high rates

Substrate, complex and transition state

- the chemical which is converted into a product is called a substrate, and enzymes, proteins with three dimensional structures including an active site, convert a substrate into a product
- the substrate has to bind the enzyme at its active site within a given cleft or pocket, diffusing towards it, so there is an obligatory formation of an enzyme substrate complex, which is a transient phenomenon
- while the substrate is bound at the active site a new structure of the substrate, called the transition state structure, is formed
- very soon after the expected bond breaking and making is completed the product is released from the active site, the structure of substrate having been transformed into that of the product, and the pathway must go through the transition state structure

The y-axis is potential energy, the x-axis the progress of reaction

Substrate and product sit at the two ends

Transition state is the peak between them

Activation energy without enzyme is the taller barrier

Activation energy with enzyme is the lower one

Figure 9.D Concept of activation energy

Intermediate states

- there could be many more altered structural states between the stable substrate and the product, and implicit in this is that all other intermediate structural states are unstable
- stability is related to the energy status of the molecule or structure

Reading the graph

- the y-axis represents the potential energy content and the x-axis the progression of the structural transformation through the transition state
- one notices the energy level difference between substrate and product, and if the product is at a lower level than the substrate the reaction is exothermic, needing no supply of energy by heating to form the product
- whether exothermic or spontaneous, or endothermic and energy requiring, the substrate has to go through a much higher energy state, the transition state
- the difference in average energy content of the substrate from that of the transition state is called activation energy, and enzymes bring down this energy barrier, making the transition easier

9.14 Nature of enzyme action

The catalytic cycle

- each enzyme has a substrate binding site in its molecule so that a highly reactive enzyme substrate complex is produced, which is short-lived and dissociates into its products and the unchanged enzyme, with an intermediate formation of the enzyme product complex
- the formation of the enzyme substrate complex is essential for catalysis
- first, the substrate binds to the active site of the enzyme, fitting into it
- the binding of the substrate induces the enzyme to alter its shape, fitting more tightly around the substrate

Effect of pH, activity peaking at an optimum

Effect of temperature, activity peaking at an optimum

Effect of substrate concentration, velocity rising to a maximum

Figure 9.E Effect of change in : (a) pH (b) Temperature and (c) Concentration of substrate on enzyme activity

The cycle, concluded

- the active site of the enzyme, now in close proximity of the substrate, breaks the chemical bonds of the substrate and the new enzyme product complex is formed
- the enzyme releases the products of the reaction and the free enzyme is ready to bind to another molecule of the substrate and run through the catalytic cycle once again

9.15 Factors affecting enzyme activity

Temperature and pH

- activity can be affected by any change in conditions that alters the tertiary structure of the protein, including temperature, pH, change in substrate concentration, or binding of specific chemicals that regulate activity
- enzymes generally function in a narrow range of temperature and pH, each showing its highest activity at a particular temperature and pH, the optimum temperature and optimum pH
- activity declines both below and above the optimum value
- low temperature preserves the enzyme in a temporarily inactive state, whereas high temperature destroys enzymatic activity because proteins are denatured by heat

Substrate concentration

- with increase in substrate concentration the velocity of the enzymatic reaction rises at first
- the reaction ultimately reaches a maximum velocity, which is not exceeded by any further rise in concentration of the substrate
- this is because the enzyme molecules are fewer than the substrate molecules, and after saturation there are no free enzyme molecules to bind the additional substrate

Inhibition

- the activity of an enzyme is also sensitive to the presence of specific chemicals that bind to it, and when such binding shuts off enzyme activity the process is called inhibition and the chemical an inhibitor
- when the inhibitor closely resembles the substrate in molecular structure and inhibits the enzyme, it is a competitive inhibitor
- due to its close structural similarity it competes with the substrate for the substrate binding site, so the substrate cannot bind and enzyme action declines, as in the inhibition of succinic dehydrogenase by malonate, which closely resembles succinate
- such competitive inhibitors are often used in the control of bacterial pathogens

9.16 Classification and nomenclature of enzymes

Six classes

- thousands of enzymes have been discovered, isolated and studied, and most have been classified by the type of reactions they catalyse
- enzymes are divided into 6 classes, each with 4 to 13 subclasses, and named accordingly by a four-digit number
- oxidoreductases or dehydrogenases catalyse oxidoreduction between two substrates
- transferases catalyse transfer of a group other than hydrogen between a pair of substrates

The remaining four

- hydrolases catalyse hydrolysis of ester, ether, peptide, glycosidic, carbon to carbon, carbon to halide or phosphorus to nitrogen bonds
- lyases catalyse removal of groups from substrates by mechanisms other than hydrolysis, leaving double bonds
- isomerases include all enzymes catalysing inter-conversion of optical, geometric or positional isomers
- ligases catalyse the linking together of 2 compounds, for instance joining of carbon to oxygen, carbon to sulphur, carbon to nitrogen and phosphorus to oxygen bonds

9.17 Co-factors

Non-protein constituents

- enzymes are composed of one or several polypeptide chains, but in a number of cases non-protein constituents called co-factors are bound to the enzyme to make it catalytically active

Three kinds of cofactor

- in these instances the protein portion of the enzyme is called the apoenzyme, and three kinds of cofactor may be identified, prosthetic groups, co-enzymes and metal ions
- prosthetic groups are organic compounds distinguished from other cofactors in being tightly bound to the apoenzyme, as haem is in peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem being part of the active site
- co-enzymes are also organic compounds but their association with the apoenzyme is only transient, usually during the course of catalysis, and they serve as co-factors in a number of different enzyme catalysed reactions, their essential chemical components often being vitamins, as niacin is in NAD and NADP
- a number of enzymes require metal ions, which form coordination bonds with side chains at the active site and at the same time with the substrate, as zinc does for carboxypeptidase

Why cofactors matter

- catalytic activity is lost when the co-factor is removed from the enzyme
- this testifies that they play a crucial role in the catalytic activity of the enzyme

9.18 Summary

Composition

- although there is a bewildering diversity of living organisms, their chemical composition and metabolic reactions appear remarkably similar, and elemental composition of living tissue and non-living matter appears similar when analysed qualitatively
- a closer examination reveals that the relative abundance of carbon, hydrogen and oxygen is higher in living systems, and the most abundant chemical in living organisms is water
- there are thousands of small molecular weight biomolecules: amino acids, monosaccharide and disaccharide sugars, fatty acids, glycerol, nucleotides, nucleosides and nitrogen bases
- there are 20 types of amino acids and 5 types of nucleotides, fats and oils are glycerides in which fatty acids are esterified to glycerol, and phospholipids contain in addition a phosphorylated nitrogenous compound

Macromolecules and enzymes

- only three types of macromolecules, proteins, nucleic acids and polysaccharides, are found in living systems, lipids separating into the macromolecular fraction because of their association with membranes
- biomacromolecules are polymers made of different building blocks, proteins being heteropolymers of amino acids and nucleic acids composed of nucleotides, with a hierarchy of structures, primary, secondary, tertiary and quaternary
- nucleic acids serve as genetic material, polysaccharides are components of cell wall in plants and fungi and of the exoskeleton of arthropods, and are also storage forms of energy
- proteins serve a variety of cellular functions, many being enzymes, some antibodies, receptors, hormones and structural proteins, with collagen the most abundant in the animal world and RuBisCO in the whole biosphere

Enzymes, in brief

- enzymes are proteins which catalyse biochemical reactions in the cells, while ribozymes are nucleic acids with catalytic power
- proteinaceous enzymes exhibit substrate specificity and require optimum temperature and pH for maximal activity, and are denatured at high temperatures
- enzymes lower activation energy of reactions and greatly enhance their rate
- nucleic acids carry hereditary information and are passed on from parental generation to progeny

9.19 Exercises

From this chapter

1. What are macromolecules? Give examples.
2. What is meant by tertiary structure of proteins?
3. Find and write down structures of 10 interesting small molecular weight biomolecules. Find if there is any industry which manufactures the compounds by isolation. Find out who are the buyers.
4. Find out and make a list of proteins used as therapeutic agents. Find other applications of proteins, for example in cosmetics.
5. Explain the composition of triglyceride.
6. Can you attempt building models of biomolecules using commercially available atomic models, the ball and stick models?
7. Draw the structure of the amino acid, alanine.
8. What are gums made of? Is Fevicol different?
9. Find out a qualitative test for proteins, fats and oils, amino acids and test any fruit juice, saliva, sweat and urine for them.
10. Find out how much cellulose is made by all the plants in the biosphere and compare it with how much paper is manufactured by man, and hence what is the consumption of plant material by man annually.
11. Describe the important properties of enzymes.